SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I G. GOWTHAMI(192524285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would

retrieve the original message.

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:
“WECRLTEERDSOEEFEAOCAIVDEN”.

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ANSWERS

1. Event Communication Security — Caesar Cipher

a. Encryption with a shift of 4

Using a Caesar cipher, each letter is shifted 4 positions forward:

- M → Q
- E → I

- E → I
- T → X
- A → E
- T → X
- N → R
- O → S
- O → S
- N → R

Therefore:
Plaintext: `MEET AT NOON`
Ciphertext: `QIIX EX RSSR`

b. Decryption

An authorized member who knows the shift is 4 shifts every encrypted letter 4 positions backward.
`QIIX EX RSSR` → `MEET AT NOON`
Thus, the original message is recovered.

2. Intercepted Suspicious Message — Atbash Cipher

a. Original message

The ciphertext is:
`GSRH RH Z HVXIVG`
The most likely original message is:
`THIS IS A SECRET`

b. Justification

This is an Atbash cipher, where the alphabet is reversed:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- ...
- G → T
- S → H
- R → I
- H → S
- Z → A
- V → E
- X → C
- I → R
- G → T

Therefore:
`GSRH` → `THIS`
`RH` → `IS`
`Z` → `A`
`HVXIVG` → `SECRET`
The word lengths and spaces also match exactly:
`4 2 1 6` → `THIS IS A SECRET`
This strongly indicates a simple substitution cipher, specifically Atbash.

3. Secure Messaging System — Vigenère Cipher

The keyword is KEY and the plaintext is HELLO.

Using A = 0, B = 1, ..., Z = 25:

a. Step-by-step encryption

Repeat the keyword until it matches the message length:

```
Plaintext:  H  E  L  L  O
Keyword:    K  E  Y  K  E
```

Convert letters to numerical values:

```
Plaintext:   H   E   L   L   O
             7   4  11  11  14

Keyword:     K   E   Y   K   E
            10   4  24  10   4
```

Add the corresponding values modulo 26:

```
7 + 10 = 17 → R
4 + 4  = 8  → I
11 + 24 = 35 → 9 → J
11 + 10 = 21 → V
14 + 4 = 18 → S
```

Therefore:

Plaintext: HELLO

Keyword: KEYKE

Ciphertext: RIJVS

b. Effect of the repeating keyword

The keyword KEY is repeated as:

KEYKEYKEY...

Each letter of the keyword determines how far the corresponding plaintext letter is shifted. Therefore, the same plaintext letter can produce different ciphertext letters depending on which keyword letter is applied.

This makes the Vigenère cipher stronger than a simple Caesar cipher because it uses multiple shifts rather than one fixed shift.

4. Suspicious Digital Asset Investigation — Steganography

a. Detecting hidden data

Investigators can examine the image for signs of steganography, where information is hidden inside an apparently normal file.

Useful approaches include:

- Visual inspection: Zoom in and look for unusual pixels, color patterns, distortions, or artifacts.
- File analysis: Examine the file size, metadata, file structure, and compression information for anomalies.
- Statistical analysis: Analyze pixel/color distributions to identify unusual patterns that may indicate embedded data.

- Least Significant Bit (LSB) analysis: Check whether the least significant bits of pixels contain patterns consistent with hidden text or files.
- Compare with the original: If an original copy exists, compare the two files using hashes and pixel-level analysis.
- Steganography detection tools: Specialized forensic software can analyze common image-steganography techniques.

b. Two techniques to strengthen hidden communication

- Encrypt the hidden information before embedding it.
  Even if investigators extract the hidden data, encryption prevents them from easily reading it.
- Use a more sophisticated embedding technique.
  Instead of simply modifying obvious pixel bits, data can be distributed carefully across image regions to make statistical detection more difficult.

A strong approach is therefore encryption + steganography: encryption protects the content, while steganography attempts to conceal its existence.

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5. Legacy Encryption — Transposition Cipher

Ciphertext:

`WECRLTEERDSOEEFEAOCAIVDEN`

a. Decrypting the message

This is consistent with a 3-rail Rail Fence transposition cipher.
Writing the ciphertext back into the 3-rail pattern and reading the characters along the zigzag produces:

`WE ARE DISCOVERED FLEE AT ONCE`

So the original message is:
WE ARE DISCOVERED. FLEE AT ONCE.

b. Why it is vulnerable

A simple transposition cipher is weak in modern communication because:

- It does not change the actual letters; it only rearranges them.
- The encryption method can often be identified from the structure of the ciphertext.
- There is a very small key space when only a few rails are used.
- An attacker can try different rail arrangements quickly.
- Modern computing makes brute-force and cryptanalysis extremely easy.
- It provides little protection against sophisticated analysis.

Therefore, a 3-rail transposition cipher should not be used to protect confidential modern communications.

c. Practical enhancement

Replace the legacy transposition cipher with a modern authenticated encryption algorithm such as AES-GCM.
This provides substantially stronger confidentiality and also helps detect whether the encrypted message has been modified.

RUBRICS FOR CALCULATION:

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understa ndi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/techno lo gies	Good understa ndi ng with proper integration of most concepts	Basic understan din g; limited or partial integration	Poor understan din g; unable to integrate concepts
Experim ent al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on
Use of Tools	20	Effective and advanced use of tools/technologi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretatio n	Basic analysis with limited insights	Poor or incorrect analysis of results
Documen tat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n